INCH-POUND

MIL-P-52746C(ME)
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SUPERSEDING
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22 May 1986

MILITARY SPECIFICATION

PUMP ASSEMBLY, FLAMMABLE LIQUID, CENTRIFUGAL, SELF-PRIMING

2-INCH, 100-GPM AT 100 FEET TOTAL DYNAMIC HEAD

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel engine-driven, frame mounted, centrifugal pump assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless other-wise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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SPECIFICATIONS

FEDERAL

VV-F-800	–	Fuel Oil, Diesel.

MILITARY

MIL-P-514	–	Plates , Identification, Instruction and Marking, Blank.
MIL-T-704	–	Treatment and Painting of Materiel.
MIL-L-2104	–	Oil, Lubricating, Internal Combustion Engine.
MIL-T-5624	–	Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-B-7883	–	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-S-7916	–	Sealing compound, Thread and Gasket, Fuel, Oil, and Water Resistant.
MIL-P-10603	–	Pumps and Pumping Units, Centrifugal, Power Driven, for Water, Packaging of.
MIL-G-10924	–	Grease, Automotive and Artillery.
MIL-P-14105	–	Paint, Heat–Resisting (for Steel Surfaces).
DOD-G-24508	–	Grease, High Performance, Multi-Purpose, (METRIC).
MIL-C-46168	–	Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	–	Coating, Aliphatic, Polyurethane, Single Component, Chemical Agent Resistant.
MIL-T-83133	–	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-H28	–	Screw-Thread Standards for Federal Service.
FED-STD-595	–	Colors used in Government Procurement.

MILITARY

MIL-STD-130	–	Identification Marking of US Military Property.
MIL-STD-810	–	Environmental Test Methods.
MIL-STD-889	–	Dissimilar Metals.
MIL-STD-1472	–	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	–	Noise Limits for Army Materiel.
MS27022	–	Coupling Half, Quick Disconnect, Cam-Locking Type, Male, External Pipe Thread, Type III.
MS27026	–	Coupling Half, Quick Disconnect, Cam-Locking Type, Female, External Pipe Thread, Type VII.

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MS27028 – Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IX.
MS27029 – Coupling Half, Quick Disconnect, Cam-Locking Type, Plug, Dust, Type X.
MS27030 – Gasket, Coupling Half, Quick Disconnect, Cam-Locking Type.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13219E3949 - Pump, Centrifugal.
13220E1127 - Cable, Ground.
TA13227E9215 - Pump Assembly, Flammable Liquid, Centrifugal , Self-Priming, 2-Inch.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: USA BELVOIR RDE CTR, ATTN SATBE FSH, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to: AMERCN SCTY FOR TEST & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

B2.1 - Welding Procedure and Performance Qualification.
D1.1 - Structural Welding Code-Steel.
D1.2 - Structural Welding Code-Aluminum.

(Application for copies should be addressed to: AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

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HYDRAULIC INSTITUTE

Rating Standards and Test Code Standards.

(Application for copies should be addressed to: HYDRLC INST, 14600 DETROIT AVE, 712 LAKEWOOD CTR N, CLEVELAND OH 44107.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC (SAE)

J 429 - Mechanical and Material Requirements for External Threaded Fasteners.

(Application for copies should be addressed to: SCTY OF AUTOMTV ENGRS, INC, 400 CMNWLTH DR, WARRENDALE PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or though libraries or other Informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pump assembly shall be in accordance with top assembly TA13227E9215 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the pump assembly. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vender to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified on the applicable drawings.

3.3.1 Materiel deterioration prevention and control. The pump assembly shall be fabricated from compatible materials, inherently

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corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encounter in any of the applicable operation and storage environment to whic the pump assembly may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used i intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are define and detailed in MIL-STD-889.

3.3.3 Cadmium plating. Except for grade 5B and grade 8B bolts, cadmium plating shall not be used.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pump assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the pump assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactur components, pieces and parts shall not be incorporated in the pump assembly.

3.4 Threads. Threads shall conform to FED-STD-H28. Prior to assembly, threads shall be coated with sealing compound conforming to MIL-S-7916.

3.5 Environmental.

3.5.1 High temperature. The pump assembly shall not be damaged by storage in an ambient temperature of +160 °F. The pump assembly shall perform as specified herein in an ambient temperature of +125 °F.

3.5.2 Low temperature. The pump assembly shall not be damaged b storage in an ambient temperature of -65 l F. The pump assembly shall perform as specified herein in an ambient temperature of -25 °F.

3.5.3 Vibration. The pump assembly shall be capable of withstanding vibration stresses without permanent damage, deformation, or performance degradation when tested as specified in 4.5.2.3.3.

3.5.4 Shock. The pump assembly in its transit condition and prepared for field use shall be capable of withstanding the shocks normally induced by loading and unloading when tested as specified in 4.5.2.3.4.

3.6 Pump. The pump shall be a 100 gallon per minute (gpm) at 10 foot total dynamic head (TDH), centrifugal, self-priming, close coupled, volatile liquid handling pump with a required net positive

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suction head (NPSH) of not more than 8 feet of water when operating at 3600 ±50 revolutions per minute (rpm) and delivering rated flow as shown on drawing 13219E3949.

3.6.1 Hydrostatic pressure. The pump shall withstand an internal hydrostatic pressure of not less than 100 pounds per square inch (psi) without leakage or permanent distortion.

Safety. All equipment hot surfaces normally above 60 “C (140 °F), including exhaust pipes exposed to contact be personnel or which create a fire hazard, shall be fully guarded or insulated. Surface temperatures induced by climatic environment are exempt from this requirement. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Exhaust or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator.

3.8 Human factors engineering The noise enclosure and any changes to the pump assembly shall conform to human factors engineering design criteria as described in MIL-STD-1472, labeling (5.5), anthropometry (5.6) , design for maintainability (5.9) , and hazards and safety (5.13) .

3.9 Noise limits. Steady–state noise produced by the pump assembly shall conform to MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3 and 5.4 when tested in accordance with MIL-STD-1474 and 4.5.2.7. Noise levels shall not exceed 85 dB(A) at the measurement locations defined as follows: Per MIL-STD-1474, 5.1.2.1.2, occasionally occupied positions shall be 39 inches horizontally from the basic envelope of the unit and a vertical distance of 48 inches above the ground, the operator’s position shall be the point 24 inches horizontally from the throttle control and 12 inches vertically above the throttle control.

3.10 Noise enclosure. The pump-engine assembly shall be contained in a permanent noise enclosure as stated herein when required to reduce the noise levels. All components of the assembly used by the operator shall be easily accessible without removal of the enclosure and without the use of tools of any kind. Accessible components shall include the starting system, the throttle control assembly, the fuel fill port, the pump case drain port, the pump priming port, the suction and discharge connections, the engine oil dipstick and the air cleaner. The throttle control assembly shall be accessible without causing the pump assembly to exceed the required noise level while operating at full load. The noise-enclosure shall be removable to allow for required field maintenance and shall include external provisions for carrying the pump assembly. Quick release latches and appropriate handles for ease of removal shall be used. If noise reducing insulation is used within the enclosure it shall be non-combustible and shall not absorb fuel and oil. All exposed surfaces of the insulation shall be protected by a perforated metal covering.

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3.10.1 Enclosure labeling. If the enclosure is provided with access panels or doors, the panel or door shall be labeled or stenciled to indicate the items accessed through it. All fluid fill and drain ports shall be labeled or stenciled to indicate the fluid filled or drained through them (i.e., "DIESEL FUEL ONLY", "ENGINE OIL DRAIN", 'ENGINE OIL FILL", "PUMP PRIME", "PUMP DRAIN") . The outlet connection shall be labeled "OUTLET". The inlet connection shall be labeled "INLET". A label stating 'CLOSE TO RUN" shall be stenciled on both sides of all access panels and doors, if possible. The enclosure shall be labeled or stenciled "4 PERSON CARRY" .

3.10.2 Oil sump temperature. The oil sump temperature of the engine shall not exceed the engine manufacturer's recommended maximum operating sump temperature when the pump is operating at full load and an ambient temperature of 125 'F.

3.11 Reliability. The specified mean-time-between-failure (MTBF) shall be 140 hours when the pump assembly is operated at rated capacity and tested as specified in 4.5.2.6.

3.12 Maintenance ratio. The pump assembly shall have a maintenance ratio of not more than 0.06. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.13 Performance.

3,13.1 Capacity. The pump assembly shall deliver liquid hydrocarbon fuels having a specific gravity of 0.740 at a rate of not less than 100 gpm against a total dynamic head of 100 feet at 3600 plus or minus 50 rpm. The pump required net positive suction head shall not exceed 8 feet of water at the required flow rate.

3.13.2 Priming. The pump shall prime in not more than 2.0 minutes.

3.14 Lubricants. The lubricants for the engine crankcase shall be as specified by the engine manufacturer. All other lubricants shall be as follows:

Lubricating Oil:	MIL-L-2104	
Grease:	MIL-G-10924	and DOD-G-24508

3.15 Engine. The engine shall be a commercial 2- or 4-stroke cycle, diesel engine. The engine shall deliver power at its continuous duty horsepower rating to meet the pump requirements specified in 3.12. The engine shall be capable of operating at rated capacity using fuel conforming to VV-F-800 or MIL-T-5624, grade JP-4 or MIL-T-83133, grade JP-8. The engine shall be equipped with but not limited to the following:

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a. One-gallon fuel tank.
b. Mechanical fuel lift pump.
c. Spark arresting muffler with guard.
d. Dry type air filter.
e. Pull-rope starting system.

3.15.1 Intermediate coupling. Intermediate coupling (adapter housing, subshaft) for interfacing to the pump shall be provided.

3.16 Frame. The pump-engine combination shall be mounted on a common frame. Individual mountings shall be furnished as required on the frame for supporting engine, pump, and other components in vertical alinement.

3.17 Connectors.

3.17.1 Inlet connector. The pump assembly inlet connector shall be a 2-inch, female, coupling half, MS27026-11, with dust plug, MS27029-11, and gasket, MS27030-6.

3.17.2 Outlet connector. The pump assembly outlet connector shall be a 2-inch, male, coupling half, MS27022-11, with dust cap, MS27028-11, and gasket, MS27030-6.

3.18 Rollover. The pump assembly shall be capable of being rolled over in any direction without sustaining cracks or permanent deformation.

3.19 Grounding assembly.

3.19.1 Grounding cable. one ground cable, conforming to drawing 13220E1127, shall be provided. Storage provisions for the grounding cable shall provided in an easily accessible compartment in the noise enclosure.

3.19.2 Ground terminal. A ground terminal shall be provided and shall be bonded to the pump assembly frame. Easy accessibility to the ground terminal shall be provided through the noise enclosure.

3.20 Cleaning, treatment and Painting. Cleaning, treatment and painting of the entire pump assembly shall be in accordance with MIL-T-704, type F or G and as stated herein.

a. All external surfaces of the pump assembly (except those that reach a temperature of 400 °F), regardless of the material selected, shall have a finish coat of CARC paint in accordance with MIL-C-46168 or MIL-C-53039. All marking and caution lettering shall be flat black. Any external surface that reaches a temperature of 400 °F during operation shall be finish coated with paint conforming to MIL-P-14105. Unless otherwise specified (see 6.2), color shall be tan 686A, color chip No. 33446 in accordance with FED-STD-595.

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b. All other surfaces of the pump assembly, to include thos within the noise enclosure and those behind insulation material, shall be treated and painted using manufacture commercial practices. The engine, pump, internal components, hardware, fasteners and fittings may be fini with the manufacturer's standard commercial paint and color, plating *or* treatment.

3.21 Identification marking. The pump assembly shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump assembly on plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 materi The plates shall be attached by screws, bolts, or rivets and centrally located on top of the noise enclosure in a location tha does not interfere with accessibility or any operator functions.

3.22 Workmanship.

3.22.1 Castings and forgings. All parts, components, and assemblies of the pump assembly which include castings and forgin shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.22.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the materi Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead shearing and sawing. All bends shall be made with controlled mea to insure uniformity of size and shape. Precaution shall be take to avoid over-heating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, a corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.22.3 Welders. Before assigning any *welder* to manual welding work covered by this specification, the contractor shall provide contracting officer with certification (see 6.7) that the-welder passed qualification tests as prescribed by the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code AWS B2.1 - Welding Procedure and Performance Qualification, AWS D - Structural Welding Code-Steel, AWS D1.2 - Structural Welding Co Aluminum.

Contractors who make only horizontal welds need not qualify welde for "all position welding". Subject to approval by the Governmen the contractor's standard welder qualification may be substituted lieu of the above codes provided that the contractor's procedure equivalent to the above codes. The contractor shall be responsib for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/AWS codes.

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3.22.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be required to enable the pumping assembly to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.188 inch unless appropriate bridging techniques are used.

3.22.5 Brazing. Brazing of steels, copper alloys and nickel alloys shall conform to MIL-B-7883.

3.22.6 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J 429 or capscrews, shall be of sufficient strength to withstand the tensile strength of the bolt. All bolts, nuts, and screws shall be correctly torqued and shall have full thread engagement. Bolts shall not protrude more than two full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, and unless disapproved by the Government, the contractor's own or any other facilities suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

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4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article pump assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article pump assembly shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance, inspection shall be conducted on each production pump assembly and shall include the examinations of 4.5.1 and the tests given in column 2 of table I. Presence of a defect or failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The pump assemblies shall be examined as specified herein for the following defects:

Critical

1. Holes in casting.

Major

102. Dimensions and tolerances not as specified (see 3.1).
103. Components missing or not as specified (see 3.1).
104. Seating surfaces not as specified (see 3.1).
105. Assembly not in proper alignment (see 3.1).
106. Materials not as specified (see 3.3).
107. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment (see 3.3.1).
108. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
109. Use of cadmium plating not documented as specified (see 3.3.4).
110. Used, rebuilt, or remanufactured components, pieces, and parts incorporated in the pump assembly (see 3.3.5).
111. Threads not as specified (see 3.4).
112. Pump not as specified (see 3.6) .
113. Noise enclosure not as specified (see 3.10).
114. Enclosure labeling not as specified (see 3.10.1).
115. Lubricants not as specified (see 3.14).
116. Engine not as specified (see 3.15).

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117. Intermediate coupling not provided (see 3.15.1).
118. Frame not as specified (see 3.16).
119. Inlet connector not as specified (see 3.17.1).
120. Outlet connector not as specified (see 3.17.2).
121. Ground cable not as specified (see 3.19.1).
122. Ground terminal not as specified (see 3.19.2).
123. Cleaning, treatment and painting not as specified (see 3.20).
124. Paint color not as specified (see 3.20).
125. Identification marking incomplete or illegible (see 3.21).
126. Castings and forgings not as specified (see 3.22.1).
127. Metal fabrication not as specified (see 3.22.2).
128. Welders not certified as specified (see 3.22.3).
129. Welding not as specified (see 3.22.4).
130. Brazing not as specified (see 3.22.5).
131. Bolted connections not as specified (see 3.22.6).

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of test code standards for centrifugal pumps of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions (barometric pressure of 29.92 inches of mercury and liquid fuel at 60 to 80 °F, having a specific gravity of 0.75 and vapor pressure of 4 psi). Water, petroleum fuels or a suitable solvent may be used as the test fluid. The test fluid temperature shall be between 60 to 80 °F.

4.5.2.2 Test schedule.

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TABLE I. Test schedule.

First article	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
		Environmental:		
X	-	High temperature	4.5.2.3.1	3.5.1, 3.10.2
X	-	Low temperature	4.5.2.3.2	3.5.2
X	-	Vibration	4.5.2.3.3	3.5.3
X	-	Shock	4.5.2.3.4	3.5.4
X	X	Hydrostatic pressure	4.5.2.4	3.6.1
X	-	Safety	4.5.2.5	3.7
X	-	Human factors	4.5.2.6	3.8
X	-	Noise level	4.5.2.7	3.9
X	-	Reliability	4.5.2.8	3.11
X	-	Maintainability	4.5.2.9	3.12
		Performance:		
X	-	Pump	4.5.2.10.1	3.13.1
X	X	Priming	4.5.2.10.2	3.13.2
X	-	Operational	4.5.2.10.3	3.13.3
X	X	Production	4.5.2.10.4	3.13.4
X	-	Rollover	4.5.2.11	3.18

4.5.2.3 Environmental.

4.5.2.3.1 High temperature The pump assembly shall be tested as specified in MIL-STD-810, method 501, procedures I and II. The maximum ambient storage temperature shall be +160 °F (+70 °C) , and the maximum operating temperature shall be +125 °F (-+52 °C). The storage period of procedure I shall be 24 hours, After temperature stabilization at 125 °F and using constant temperature exposure, the operating period of procedure II shall be 3 hours. The pump assembly shall be operated at rated capacity. The oil sump temperature of the engine shall be measured during the operating period of procedure II. Nonconformance to 3.5.1 and 3.10.2 shall constitute failure of this test.

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4.5.2.3.2 Low temperature. The pump assembly shall be tested as specified in MIL-STD-810, method 502, procedures I and II. The minimum ambient storage temperature shall be -65 °F (-54 °C), and the minimum ambient operating temperature shall be -25 °F (–32 °C) The storage period of procedure I shall be 24 hours. After temperature stabilization at -25 °F and using constant temperature exposure, the operating period of procedure II shall be 3 hours. The pump assembly shall be operated at rated capacity. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.3.3 Vibration. The pump assembly, unpackaged, shall be subjected to the tests specified in MIL-STD-810, method 514, procedure III, equipment category 3. Any breakage, permanent deformation, or other damage to the pump assembly or failure of the pump assembly to operate as specified shall constitute failure of this test.

4.5.2.3.4 Shock. The pump assembly shall be subjected to the tests specified in MIL-STD-810, method 516, procedure IV. The pump assembly shall be dropped on the bottom corners only for a total of four drops. Any breakage, permanent deformation, or other damage to the pump assembly or failure of the pump assembly to operate as specified shall constitute failure of this test.

4.5.2.4 Hydrostatic pressure. Subject the pump to an internal hydrostatic pressure of not less than 100 psi for not less than 5 minutes. Leakage from any part of the pump, cracks in the pump, or permanent deformation of any component shall constitute failure of this test.

4.5.2.5 Safety. The pumping assembly shall be evaluated throughout the testing as specified in 4.5.2. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.6 Human factors. The human factors engineering requirements shall be evaluated during testing. Nonconformance to 3.8 shall constitute failure of this test.

4.5.2.7 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 11. As a minimum, noise levels shall be measured when the equipment is operating under full load. Noise levels shall be measured at not fewer than 12 equal (horizontal) arc increments for the occasionally occupied positions. Additionally, the noise level at the typical operating position (see 3.8) shall be provided as dB(A) level. Failure to comply with 3.9 provisions shall constitute failure of this test.

4.5.2.8 Reliability. The first article pump assembly shall be operated at its rated capacity under the standard conditions specified in 4.5.2.1. The operational test specified in 4.5.2.8.3 shall constitute the first 100 hours of this test. The pump assembly shall be continuously operated for 9 hours out of every 24 hours. Normal maintenance shall be performed in accordance with the

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manufacturer's operation and maintenance schedule. A failure is defined as any malfunction which:

a. Cannot be corrected within 30 minutes by adjustment, repair or replacement using only the maintenance tools and repair parts furnished with the equipment; or
b. May cause failure to commence operation, cessation of operation, or degradation of performance below specified level; or
c. May damage pump assembly by continued operation; or
d. May cause a safety hazard to operating personnel.

The total test time is total unit hours of pump "on" time and is expressed in multiples of specified MTBF (see 3.10). The test shall continue for a sufficient length of time to reach a decision to "accept" or "reject" in accordance with figure 1. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.9 Maintainability.

4.5.2.9.1 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.12 shall constitute failure of this test.

4.5.2.10 Performance .

4.5.2.10.1 Pump. With the pump operating at 3600 rpm and a static suction lift equivalent to 13 feet of water, test the pump in accordance with the Rating Standards and Test Code Standards for centrifugal pumps of the Hydraulic Institute. Data relative to head-capacity shall be taken at a minimum of 6 points and shall include the rating point, wide open discharge and shutoff. The pump assembly shall run for a minimum of 5 minutes at each measured point. Delivery of less than the capacity specified in 3.13.1 or NPSH greater than specified in 3.13.1 or failure of the engine to meet any of the operational requirements or conditions specified in 3.13 shall constitute failure of this test.

4.5.2.10.2 Priming. Fill the pump case with either liquid petroleum fuel or a suitable solvent having a specific gravity of 0.72 to 0.75 and conduct the priming test with the pump operating at 3600 ±50 rpm and a static suction lift equivalent to 13 feet of water at standard atmospheric conditions. Failure of the pump to prime in not more than 2 minutes shall constitute failure of this test.

4.5.2.10.3 Operational. Fill the pump case with test fluid and prime the pump assembly with a static suction lift equivalent to 13 feet of water at standard atmospheric conditions. Once the pump has primed, operate the pump at 100 gpm at 100 feet TDH for 100 hours. This test shall be performed for not less than 9 continuous hours daily until the conclusion of the test. During operation, only minor adjustments to the engine may be performed. A maximum shutdown time of 15 minutes is allowable for engine maintenance and

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examination during each 9-hour period. During operation, determinations shall be made as to whether there is leakage through the pump case, piping, fittings, or at the pump shaft seal and whether there is malfunction of any component. Leakage, malfunction of any component, inability of the pump assembly to prime in 2 minutes and deliver the capacity specified, or inability of the pump assembly to operate continuously during periods of operation shall constitute failure of this test.

4.5.2.10.4 Production- Test the pump assembly as specified in 4.5.2.10.3 except that the operating time shall be 15 minutes.

4.5.2.11 Rollover. The pump assembly shall be balanced on one longitudinal edge and then allowed to fall over. The assembly shall then be inspected for damage or broken components. Nonconformance to 3.18 shall constitute failure of this test. Nicks, dents, scrapes, and marred finished shall not be considered as defects.

4.6 Inspection of packaging. The preservation, packing and marking shall be examined and tested in accordance with the applicable quality assurance provisions of MIL-P-10603 except that commercial preservation, packing and marking shall be examined for compliance with ASTM D 3951 and the requirements of section 5 herein.

5. PACKAGING

5.1 First artitle pack. Unless otherwise specified (see 6.2) , the contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2), in accordance with the preproduction pack requirements of MIL-P-10603.

5.2 Preservation, packing, and marking. Preservation, packing, and marking shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Level A preservation, packing, and marking shall be *in* accordance with the applicable requirements of MIL-P-10603.

5.2.2 Commercial. Commercial preservation, packing, and marking shall be in accordance with ASTM D 3951. Additionally, each shipping container shall be marked with the cube and gross weight.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pump assembly is intended for use in dispensing liquid hydrocarbon fuels.

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6.2 Acquisition requirements. Acquisition documents should specify the following;

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article and number of units required (see 3.2).
d. Color of top coat if other than specified (see 3.20).
e. When a first article pack is not required (see 5.1).
f. Time frame for submission of first article pack, when required (see 5.1).
g. Degree of preservation and packing required (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Noise limits. Where the limit of Category D can be documented as being clearly beyond the state-of-the-art, per MIL-STD-2474, 5.1.1.2, selection of another noise limit should be considered by the procuring activity, as described in MIL-STD-1474, acceptance requirements (5.1.1.3) . Particular attention should be given to 5.1.1.2.2.d. When permission is granted to exceed the limits of Category D, the procuring activity should ensure that operator and maintenance manuals are prepared in accordance with MIL-STD-1474, manuals requirements (4.3).

6.5 Subject term (key word) listing.

Pump, diesel-engine-driven
Pump, centrifugal, self-priming
Fuel dispensing, 100 GPM

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure

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that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.22.3	DI-MISC-80876	Welding Procedure Qualification Test Report
3.22.3	DI-MISC-80875	Welding Procedures

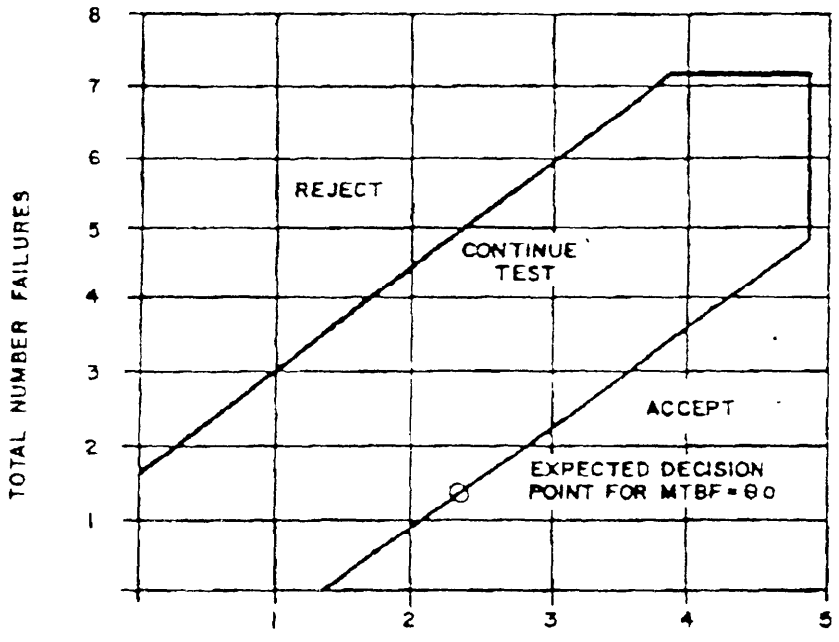
The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on *the* DD Form 1423.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4320-A346

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TOTAL TEST TIME *

NO. OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

* TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 1 Accept-reject criteria.

X-3499

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE.	1. DOCUMENT NUMBER MIL-P-52746C(ME)	2. DOCUMENT DATE (YYMMDD) 940620

3. DOCUMENT TITLE
PUMP ASSEMBLY, FLAMMABLE LIQUID, CENTRIFUGAL, SELF-PRIMING 2-INCH, 100-GPM AT 100 FEET TOTAL DYNAMIC HEAD

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) DSN 654-3468
c. ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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